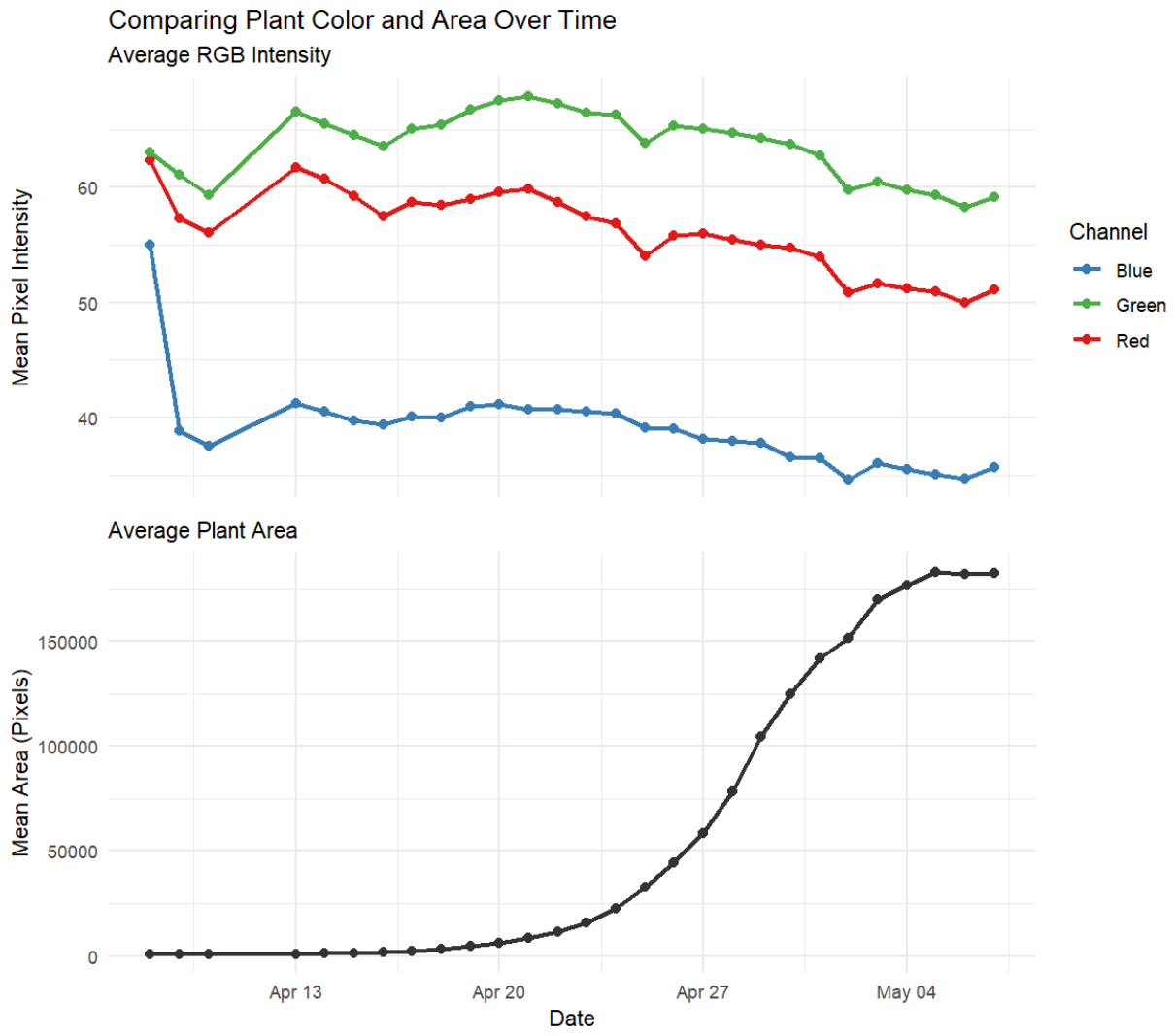
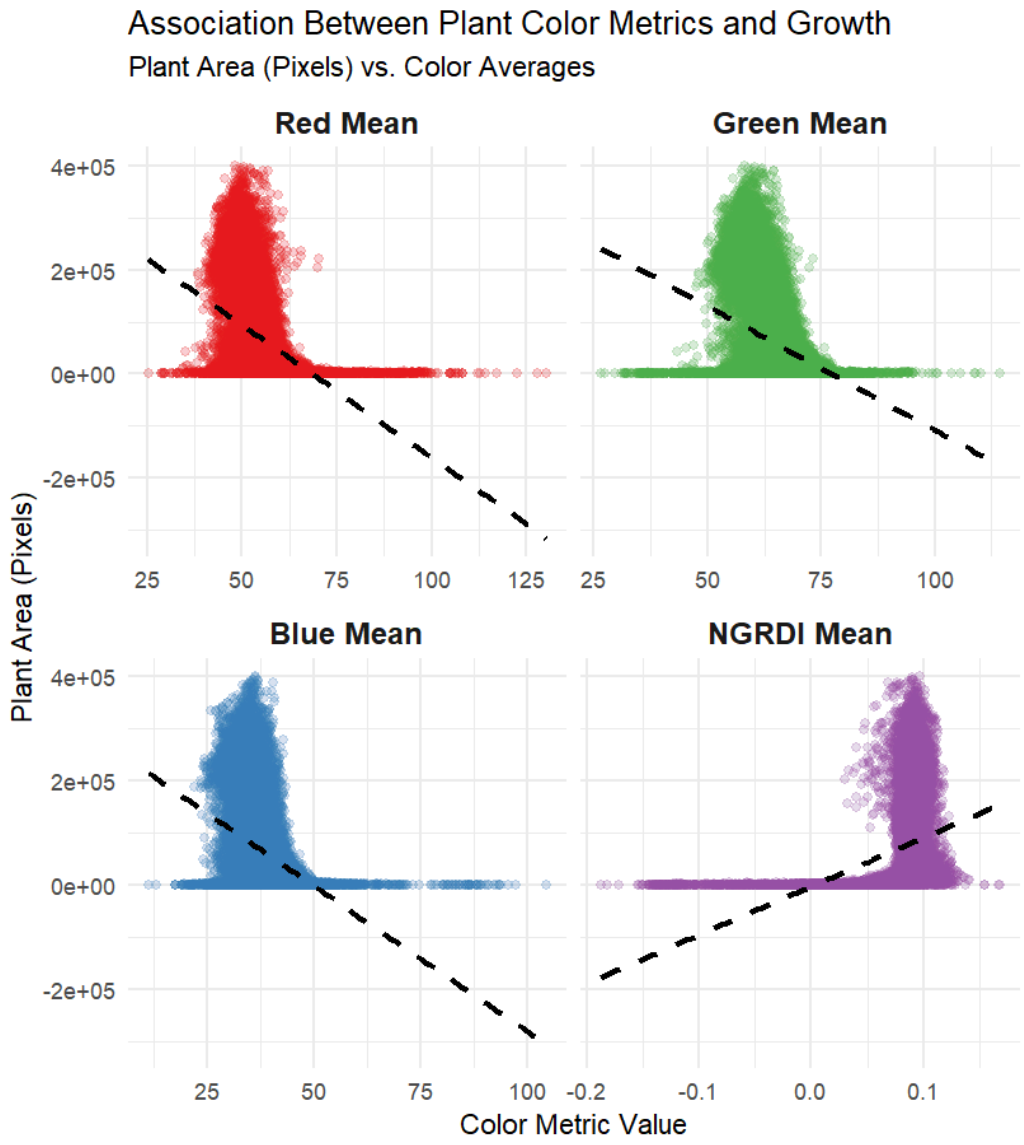


What I did:

I used the date column to sort the data by days, and took the mean pixel intensity for each color per day. I then did the same thing for the mean of the areas for each day.

Found:

Found out that as the plant grows, the intensity of each of the colors decreases. I believe this to be due to shadowing as the plant grows. The leaves above shadow the ones below, lowering the intensity of the color.

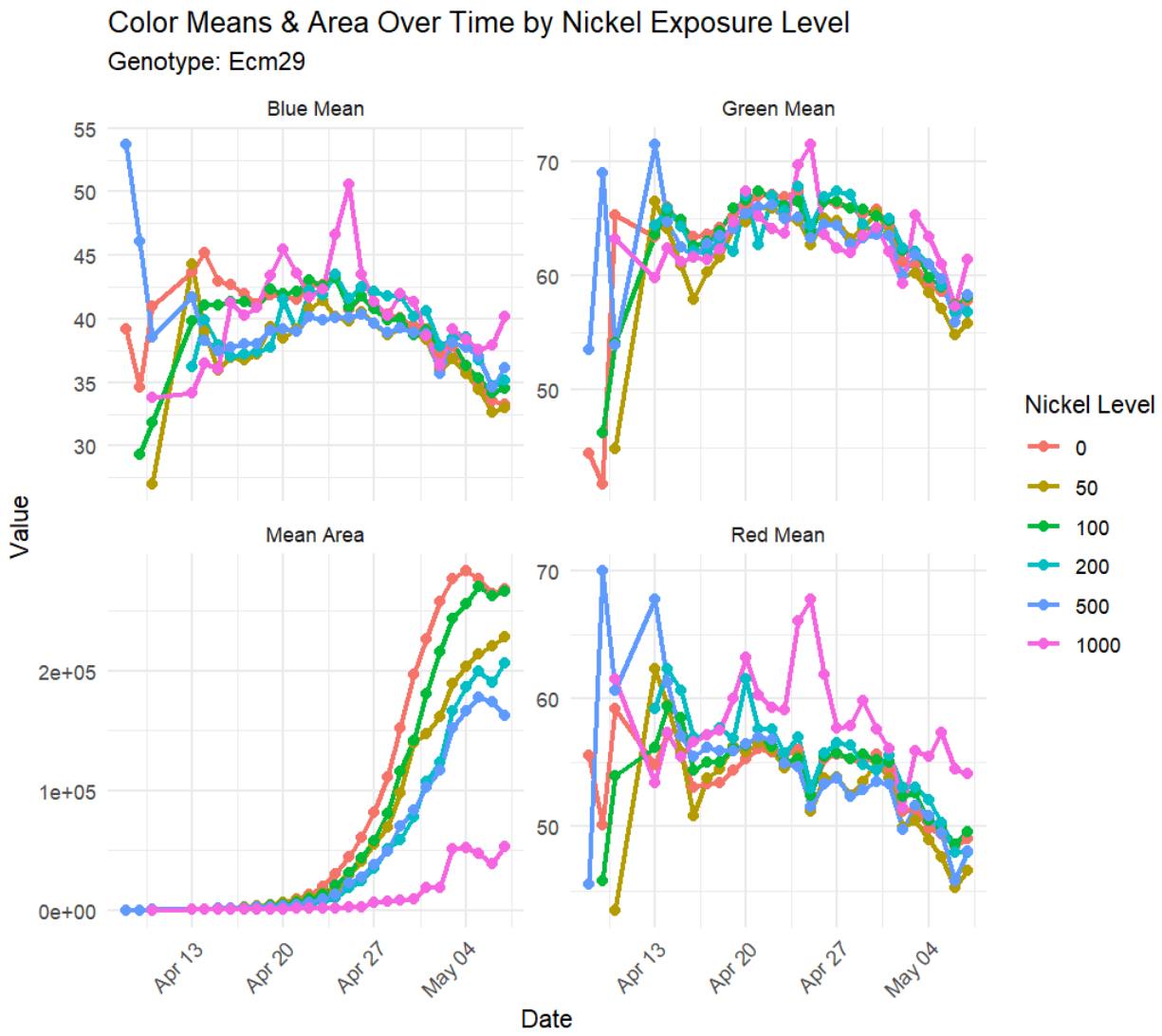


What I did:

Took a scatter plot of the area of the plant versus the pixel intensity value for each color mean as well as the NGRDI. I also added an OLS regression line for each to give an idea on the general trend.

Found:

Not a whole lot, besides confirming the idea from the previous plot that the colors are negatively related to the area of the plant. It did tell me that the NGRDI is positively related, and based on the slope, it looks like it is slightly less related than the color means.

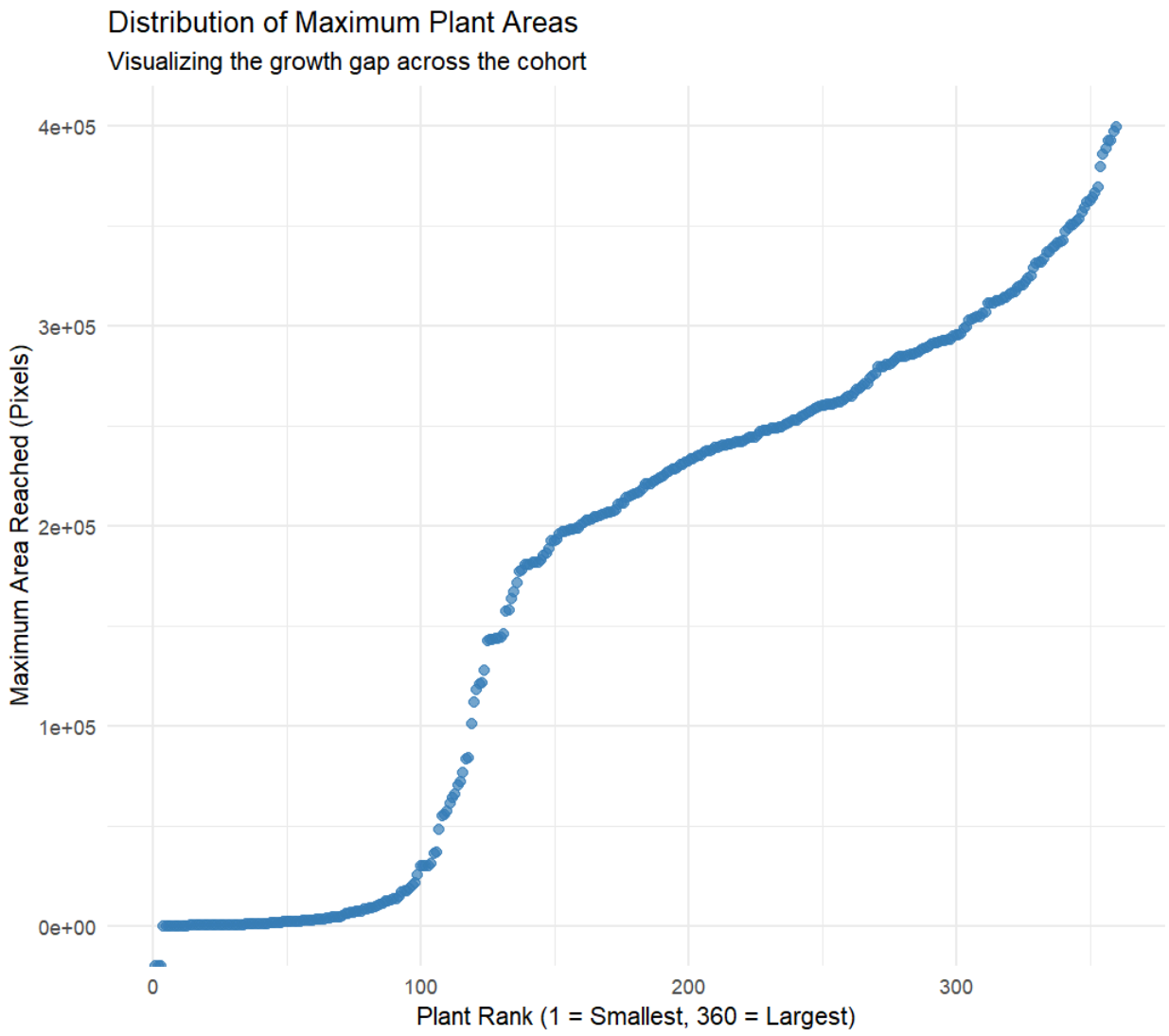


What I did:

Made a separate graph for each genotype. I then made the treatments each their own colored line, and based on those two filters, graphed each color and the area over time. I do not see a point in posting each genotype unless you want them, so just let me know if you want all of the genotypes. It is also in the github code under section “MEAN COLOR AND AREA BY TREATMENT AND GENOTYPE”

Found:

Each treatment has a very different curve for the color, and some variation in the later days for area. However, the treatments stay relatively clustered together. With some breaking off for the extreme treatments

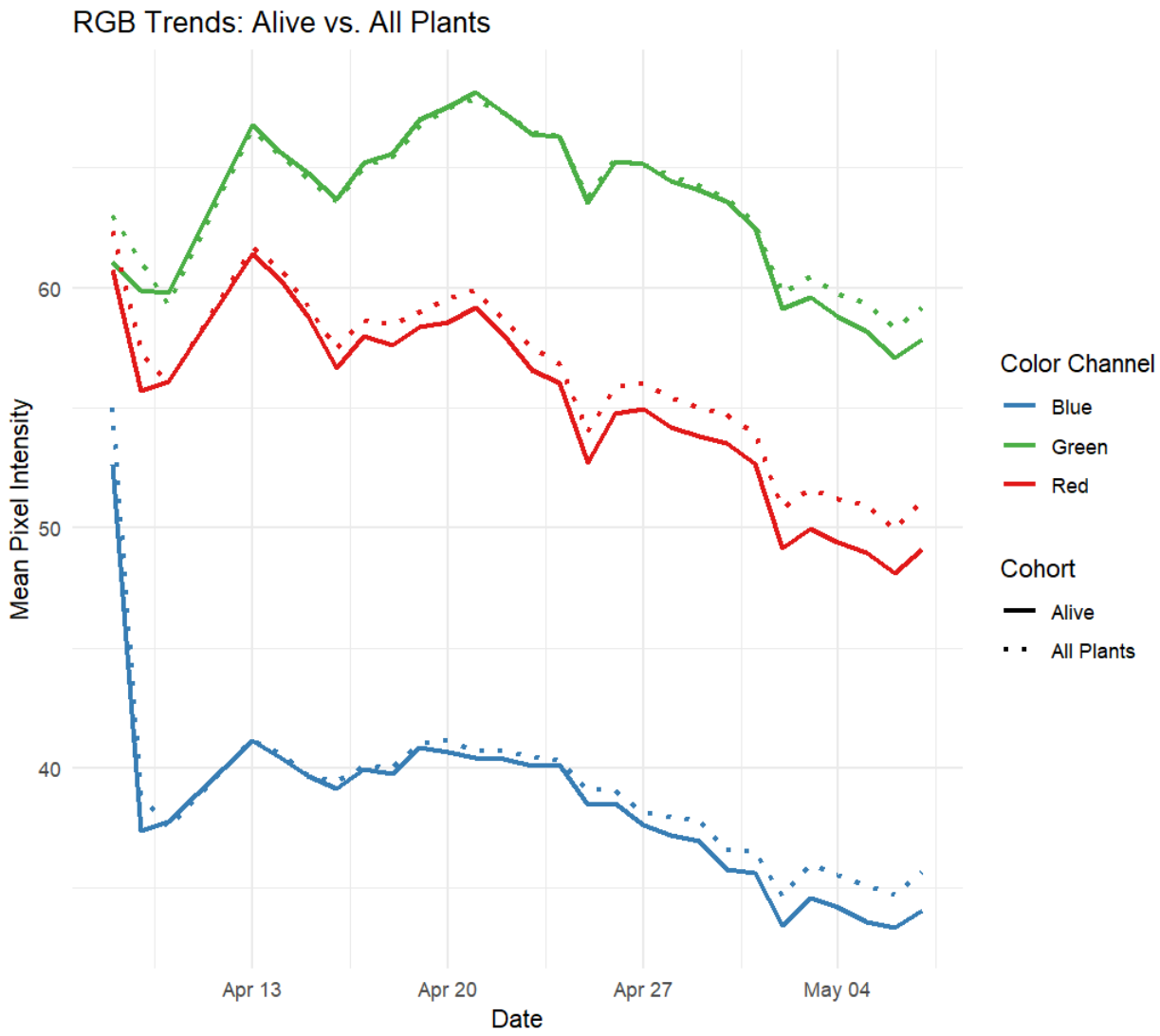


What I did:

Took the max area a specific plant reached and put them from smallest to largest, and graphed it as a scatter plot

Found:

There is a gap between rank 100 and 150. This seems to break the plants into two groups. I considered anything below rank 100 a dead plant since they never grew above 20,000 pixels. Allowed me to compare alive and dead plants

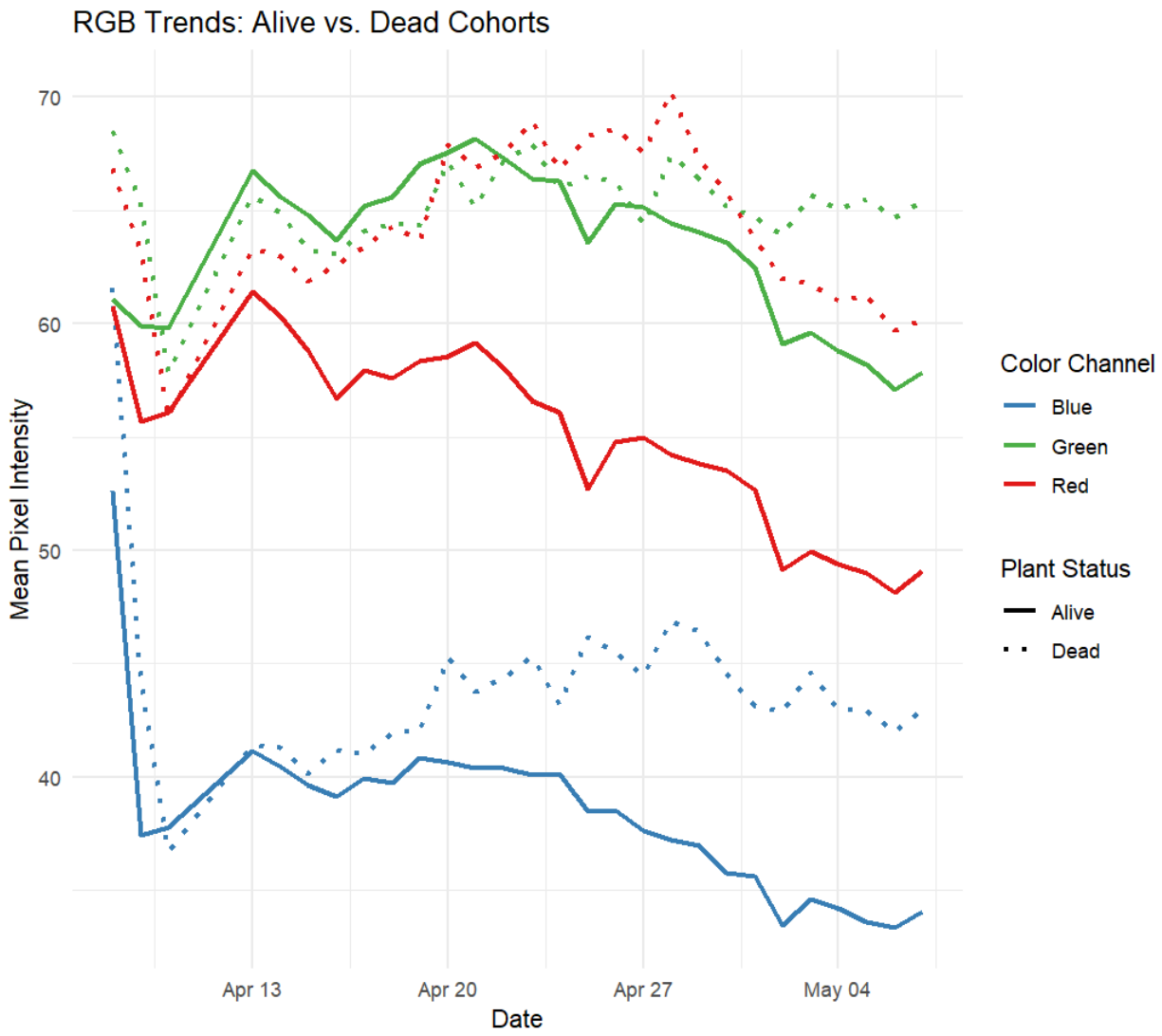


What I did:

Using the 20,000 pixel threshold from above, I separated the dataset into alive versus dead plants. I then took the mean pixel intensity over time for the alive plants and all the plants and overlaid them for easy comparison.

Found:

The red line for all plants is strictly above the line for alive plants after day 5 (April 13th). Same for the blue line, just much less noticeable. The green line seems to stay around the same.



What I did:

Same thing as the graph above, only difference is an overlaid alive and dead rather than alive and all.

Found:

This showed a bigger difference. This could be simply due to the alive versus dead difference. However, the dead plants only had 96 plants but the alive ones had 264. However, this graph shows that after April 13th (Day 5), the red and blue values for dead plants stay above the values for alive plants. Green however, oscillates around the alive plant values.

```
[1] "--- Summary Statistics: WHOLE DATASET ---"
> df_classified %>%
+   select(red_mean, green_mean, blue_mean) %>%
+   summary() %>%
+   print()
    red_mean        green_mean       blue_mean
 Min.   : 25.30   Min.   : 26.50   Min.   : 11.29
 1st Qu.: 51.01   1st Qu.: 59.78   1st Qu.: 34.65
 Median : 55.65   Median : 64.47   Median : 38.33
 Mean   : 56.20   Mean   : 64.18   Mean   : 38.68
 3rd Qu.: 59.75   3rd Qu.: 68.49   3rd Qu.: 42.21
 Max.   :130.37   Max.   :114.36   Max.   :104.48
 NA's   :4690     NA's   :4690     NA's   :4690
> # 4. Generate summary statistics for ALIVE plants only
> print("--- Summary Statistics: ALIVE PLANTS ---")
[1] "--- Summary Statistics: ALIVE PLANTS ---"
> df_classified %>%
+   filter(status == "Alive") %>%
+   select(red_mean, green_mean, blue_mean) %>%
+   summary() %>%
+   print()
    red_mean       green_mean      blue_mean
 Min.   :28.66   Min.   :31.16   Min.   :17.68
 1st Qu.:50.69   1st Qu.:59.89   1st Qu.:34.40
 Median :55.17   Median :64.37   Median :37.93
 Mean   :55.16   Mean   :64.07   Mean   :38.09
 3rd Qu.:58.94   3rd Qu.:68.27   3rd Qu.:41.72
 Max.   :98.10   Max.   :98.75   Max.   :83.80
 NA's   :629     NA's   :629     NA's   :629
> # 5. Generate summary statistics for DEAD plants only
> print("--- Summary Statistics: DEAD PLANTS ---")
[1] "--- Summary Statistics: DEAD PLANTS ---"
> df_classified %>%
+   filter(status == "Dead") %>%
+   select(red_mean, green_mean, blue_mean) %>%
+   summary() %>%
+   print()
    red_mean        green_mean       blue_mean
 Min.   : 25.30   Min.   : 26.50   Min.   : 11.29
 1st Qu.: 55.21   1st Qu.: 58.71   1st Qu.: 37.65
 Median : 61.90   Median : 65.36   Median : 41.62
 Mean   : 63.97   Mean   : 65.05   Mean   : 43.08
 3rd Qu.: 70.55   3rd Qu.: 71.02   3rd Qu.: 46.62
 Max.   :130.37   Max.   :114.36   Max.   :104.48
 NA's   :4061     NA's   :4061     NA's   :4061
```

Difference between min & max

Red: 105.07
Green: 87.86
Blue: 93.19

Red: 69.44
Green: 67.59
Blue: 66.12

Red: 105.07
Green: 87.86
Blue: 93.19

What I did:

I took the split up datasets from above and ran some summary statistics on them. I also calculated the difference between the min and max value and recorded that on the side. These summary statistics are based on individual observations, not averages. So, the max red value of 130.37 is the highest value a single plant reached.

Found: First, I noticed that the difference between the min and the max of rgb values was exactly the same for the dead plants as well as the whole dataset. Additionally, I noticed that the differences are much lower for the alive plants dataset. The differences between the rgb values are also much closer together between each color for the alive plants with a difference between red, green and blue only being about 4. Whereas the difference between red, green and blue for the dead plants is about 20. Through a little bit of research, I learned that the likely reason the living plants have such a tighter range is due to the chlorophyll regulation. When a plant is alive, it can accurately regulate its chlorophyll to allow for the proper amount of sunlight to be captured. However, when a plant is dying or dead, it can no longer regulate its colors, allowing for a larger range. I thought it was odd that the dead plants have both the largest and the smallest numbers, so I had looked into that as well. The high values are likely due to bleaching or drying out when the plant is dying. The low values are likely due to the plant not growing much, so there are not many pixels to average out, or necrosis taking over.

0	50	100	200	500	1000
11	11	10	5	27	32

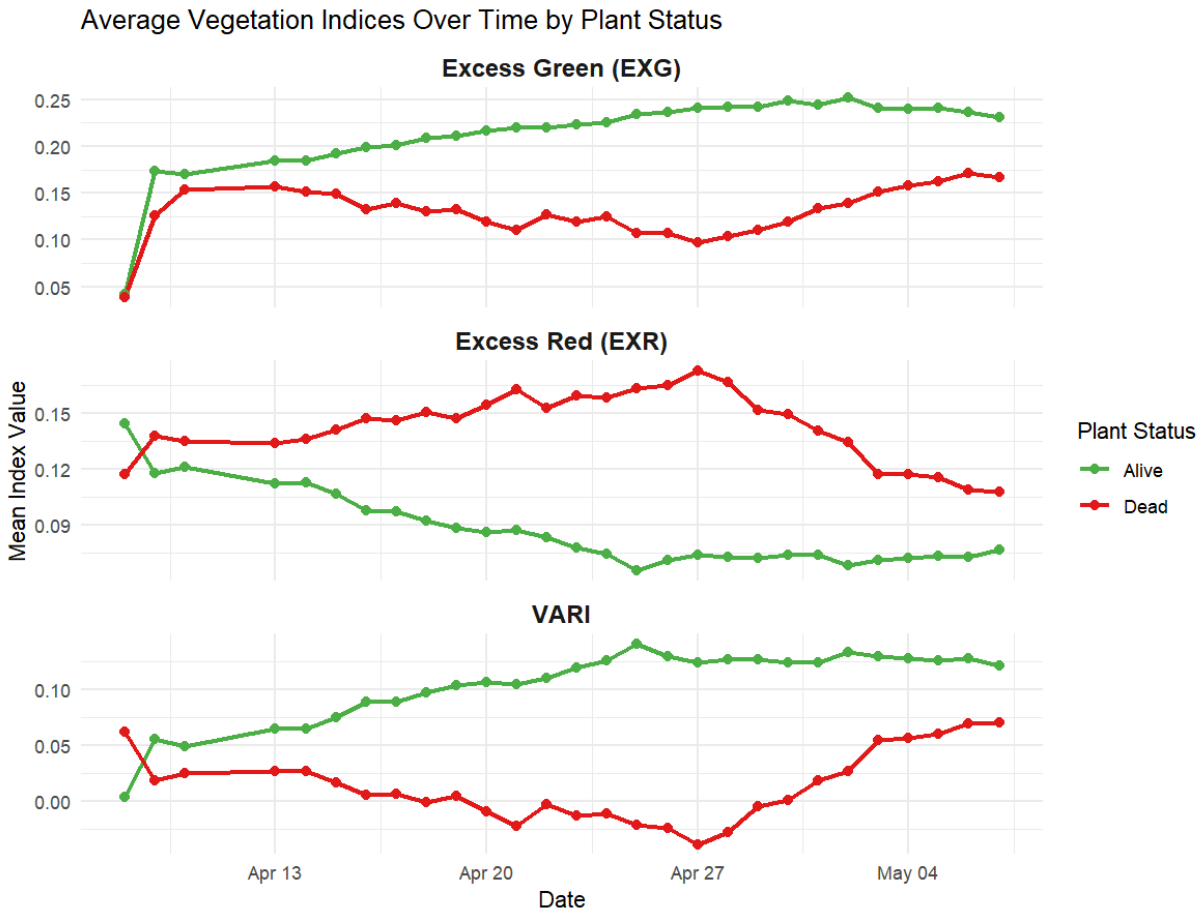
What I did:

I took the dead plants dataset and looked at how many died for each treatment level.

Found:

Interestingly, less plants died with 200 nickel exposure than with 0, 50 or 100 nickel exposure. Unsurprisingly, the most plants died from the 500 and the 1000 nickel exposure.

Note, also found that there are 360 individual plants in the experiment, which is over a 29 day window between 04-08-2026 and 05-07-2026



What I did:

In the research paper for the lettuce, it mentioned the EXG, EXR and VARI values. I calculated these and graphed their daily means by whether I marked them as alive or dead in the previous datasets.

Note, EXG is chlorophyll concentration, EXR is for detecting vegetative stress, and VARI normalizes the shadowing effect, showing the true color concentration

Found:

The EXG is significantly higher for the living plants than the dead ones. The difference starts getting even wider by Apr 13. The dead plants do increase in concentration towards the end of the month, but never pass the alive plants.

The EXR is significantly lower for the living plants than the dead ones. This makes sense since the EXR shows vegetative stress. It is interesting that they start out with the living plants higher than the dead ones, but after a day or two, it switches and they stay with the dead ones higher. By Apr 14-15, the values start separating even further.

The VARI is significantly higher for the living plants than the dead ones. The difference gets increasingly wider starting around Apr 14-15. Just like with the EXR, the values start with the dead plants being higher than the living ones, but switches after a day or two and stays with the living plants higher.
All of the lines start close together, get significantly further apart, and then get close again towards the final week. The lines never cross past day 2 or 3.

```
Call:
glm(formula = is_alive ~ early_EXG + early_EXR + early_VARI +
    early_NGRDI, family = "binomial", data = early_color_predictors)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)    7.519      9.913   0.759    0.448
early_EXG      7.354      7.388   0.995    0.319
early_EXR    -55.121     71.322  -0.773    0.440
early_VARI   -46.036     26.295  -1.751    0.080 .
early_NGRDI   46.452     66.124   0.703    0.482
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 358.75  on 325  degrees of freedom
Residual deviance: 314.39  on 321  degrees of freedom
AIC: 324.39

Number of Fisher Scoring iterations: 5
```

What I did:
Ran a logistic regression model with EXG, EXR, VARI and NGRDI as predictors to see if any of these indices can help us predict if a plant will survive.

Found:
None of the indices showed they were statistically significant. This was odd considering every graph before showed us they were significant.

```
[1] "--- Logistic Regression: Survival Predicted by Early VARI
> summary(vari_only_glm)

Call:
glm(formula = is_alive ~ early_VARI, family = "binomial", data =
color_predictors)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   0.3564     0.1942   1.836   0.0664 .
early_VARI   19.0775     3.6360   5.247 1.55e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 358.75  on 325  degrees of freedom
Residual deviance: 324.28  on 324  degrees of freedom
AIC: 328.28

Number of Fisher Scoring iterations: 4

> # 2. Test NGRDI purely on its own
> ngrdi_only_glm <- glm(is_alive ~ early_NGRDI,
+                       data = early_color_predictors,
+                       family = "binomial")
>
> print("--- Logistic Regression: Survival Predicted by Early N
")
[1] "--- Logistic Regression: Survival Predicted by Early NGRDI
> summary(ngrdi_only_glm)

Call:
glm(formula = is_alive ~ early_NGRDI, family = "binomial", data
_color_predictors)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   0.2778     0.1987   1.398    0.162
early_NGRDI  31.3837     5.6553   5.549 2.87e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 358.75  on 325  degrees of freedom
Residual deviance: 319.92  on 324  degrees of freedom
AIC: 323.92

Number of Fisher Scoring iterations: 4
```

```
[1] "--- Logistic Regression: Survival Predicted by Early EXG
> summary(exg_only_glm)

Call:
glm(formula = is_alive ~ early_EXG, family = "binomial", data
olor_predictors)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -1.3291     0.4986  -2.666  0.00768 **
early_EXG    15.4159     3.0490   5.056 4.28e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 358.75  on 325  degrees of freedom
Residual deviance: 329.22  on 324  degrees of freedom
AIC: 333.22

Number of Fisher Scoring iterations: 4

>
> # 2. Test EXR purely on its own
> exr_only_glm <- glm(is_alive ~ early_EXR,
+                     data = early_color_predictors,
+                     family = "binomial")
>
> print("--- Logistic Regression: Survival Predicted by Early
[1] "--- Logistic Regression: Survival Predicted by Early EXR
> summary(exr_only_glm)

Call:
glm(formula = is_alive ~ early_EXR, family = "binomial", data
olor_predictors)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   5.4813     0.8381   6.540 6.14e-11 ***
early_EXR   -34.4525     6.4427  -5.348 8.91e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 358.75  on 325  degrees of freedom
Residual deviance: 323.37  on 324  degrees of freedom
AIC: 327.37

Number of Fisher Scoring iterations: 4
```

What I did:

Ran the same logistic regression model as above, but with each index ran individually.

Found:

Every single index showed high significance. This told me that there is multicolinearity present, which makes sense since each of the indexes are calculated from the same averages of pixel intensity. EXR was the only index that was negatively related, but that makes sense since the living plants showed a decrease in EXR as time went on, but an increase for the other 3 indices.

```
[1] "--- Linear Mixed Model: EXG (Greenness) Trajectory ---"
> summary(exg_lmm)
Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest'
Formula: EXG ~ day_numeric * status + (1 | plant_id)
   Data: df_color_lmm

REML criterion at convergence: -78778.1

Scaled residuals:
    Min      1Q  Median      3Q     Max
-7.5798 -0.4821  0.0113  0.5388  7.7394

Random effects:
 Groups   Name        Variance  Std.Dev.
 plant_id (Intercept) 0.0007412 0.02722
 Residual             0.0008412 0.02900
Number of obs: 18890, groups:  plant_id, 357

Fixed effects:
                          Estimate Std. Error         df t value Pr(>|t|)
(Intercept)             1.791e-01  1.775e-03  3.859e+02  100.95   <2e-16 ***
day_numeric             2.685e-03  3.271e-05  1.852e+04   82.09   <2e-16 ***
statusDead             -5.105e-02  3.742e-03  5.005e+02  -13.64   <2e-16 ***
day_numeric:statusDead -2.539e-03  9.343e-05  1.888e+04  -27.18   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Correlation of Fixed Effects:
            (Intr) dy_nmr sttsDd
day_numeric -0.303
statusDead  -0.474  0.144
dy_nmrc:stD  0.106 -0.350 -0.391
> anova(exg_lmm)
Type III Analysis of Variance Table with Satterthwaite's method
                    Sum Sq Mean Sq NumDF   DenDF F value    Pr(>F)
day_numeric        0.77221 0.77221     1 18881.5  917.93 < 2.2e-16 ***
status             0.15661 0.15661     1   500.5  186.16 < 2.2e-16 ***
day_numeric:status 0.62150 0.62150     1 18881.5  738.78 < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

What I did:

Ran a linear mixed model on EXG with “Alive” status as the baseline. This model is used for non-independent measurements by adding fixed and random effects. Fixed effects show the average relationship or trend across the population and random effects account for variation in the groups or individual differences. In this case, the fixed are the

Found:

The EXG index was highly dependent on whether they lived or died. Each group had statistically different values for the EXG index. Under the fixed effects estimate, it was 0.00268, indicating the living plants increased their EXG index by

about 0.00268 each day. Additionally, the statusDead estimate shows that these plants started out 0.051 units lower in EXG index than the living plants did. The estimated change per day for the dead plants was -0.00253 units different from the living plants. Which means that the daily increase in EXG value for the dead plant was 0.00015 per day.

```
[1] "--- Linear Mixed Model: NGRDI Trajectory ---"
> summary(NGRDI_lmm)
Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLme
rTest']
Formula: ngrdi_mean ~ day_numeric * status + (1 | plant_id)
   Data: df_color_lmm

REML criterion at convergence: -90993.1

Scaled residuals:
    Min      1Q  Median      3Q     Max
-8.9414 -0.4644  0.0526  0.5266  7.8604

Random effects:
 Groups   Name        Variance  Std.Dev.
 plant_id (Intercept) 0.0004375 0.02092
 Residual             0.0004396 0.02097
Number of obs: 18890, groups:  plant_id, 357

Fixed effects:
                          Estimate Std. Error         df t value Pr(>|t|)
(Intercept)             4.198e-02  1.355e-03  3.825e+02   30.99   <2e-16 ***
day_numeric             2.123e-03  2.364e-05  1.852e+04   89.81   <2e-16 ***
statusDead             -3.932e-02  2.839e-03  4.858e+02  -13.85   <2e-16 ***
day_numeric:statusDead -1.545e-03  6.762e-05  1.889e+04  -22.85   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Correlation of Fixed Effects:
            (Intr) dy_nmr sttsDd
day_numeric -0.287
statusDead  -0.477  0.137
dy_nmrc:stD  0.100 -0.350 -0.373
> anova(NGRDI_lmm)
Type III Analysis of Variance Table with Satterthwaite's method
                   Sum Sq Mean Sq NumDF   DenDF F value    Pr(>F)
day_numeric        0.70207 0.70207     1 18885.9 1596.99 < 2.2e-16 ***
status             0.08437 0.08437     1   485.8  191.91 < 2.2e-16 ***
day_numeric:status 0.22946 0.22946     1 18885.9  521.96 < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

What I did:

The same LMM from before, but ran it for NGRDI instead of EXG.

Found:

The intercept estimate shows that the living plants started at a NGRDI of 0.04198, indicating they started out green. (We know this because based on the NGRDI formula, a positive number is more green than red, and negative is more red than green. Close to zero means green and red are about the same) The dead plants started 0.03932 units lower than the living plant did, meaning the dead plants started at a NGRDI of 0.00266. Which is significantly close to 0, indicating that the plants that ended up dying started out stressed, causing them to be equally green and red. This can mean that NGRDI can help us predict whether or not a plant will die early on. The high f-value for day_numeric shows that NGRDI is a good predictor for tracking healthy growth over time. The day_numeric:statusDead in the fixed effects tells us that the dead plants grew 0.001545 slower than the living plants, which results in a 0.000578 growth rate for the dead plants rather than the 0.002123 growth rate of the living plants.